

WHEN DOES REFERENCE MATCHING TRANSFER? SPECIFICATION-CERTIFIED AUDITS OF FIR/IIR REALIZATIONS

Xianghui Meng¹ and Jionghao Lin^{1,2,*}

¹The University of Hong Kong

²Carnegie Mellon University

margretmeng1020@gmail.com

*Corresponding author: jionghao@hku.hk

ABSTRACT

Reference-based correctness scoring treats proximity to selected realizations as a proxy for satisfaction of a digital-filter specification, even when a magnitude mask admits many legitimate FIR or IIR designs. We audit that proxy on 20 magnitude-mask tasks (16 FIR, 4 IIR) whose 412 constructed valid implementations are continuously certified, using coefficient and magnitude-response catalogs of observed valid realizations. No observed valid coefficient reference recovers exact single-center membership on any task; exact coefficient catalogs require more than 10 references on all 20 tasks (median 23). After those catalogs were frozen, a catalog-blind protocol produced 614 new continuously certified valid realizations. The frozen coefficient catalogs accepted 66/614 (10.7%; task-macro median 4.8%), whereas the frozen magnitude-response catalogs accepted 585/614 (95.3%; task-macro median 100%). Reference adequacy therefore depends on representation and on the diversity of valid realizations that must be supported: coefficient matching is tied to realization identity, while magnitude-response matching is a substantially stronger—but not universally exact—surrogate on this suite.

Index Terms— Digital filters, FIR, IIR, specification testing, reference matching

1. INTRODUCTION

A digital-filter task is specified by passband and stopband edges, ripple, attenuation, and, for IIR filters, stability [1, 2, 3]: a feasible set, not a unique coefficient vector. Evaluation nevertheless often scores proximity to a golden coefficient vector, a reference magnitude response, or a small library catalog. That surrogate asks whether a candidate is the *same realization*, not whether it satisfies the mask. This paper audits whether such realization-reference oracles transfer to alternative specification-valid FIR/IIR designs.

That Remez, window, least-squares, frequency-sampling, Butterworth, Chebyshev, and elliptic constructions can meet

one mask with different taps or sections is classical filter theory [4, 5, 6, 7, 8, 9, 10, 11, 12, 13] and is not our contribution. The general test-oracle problem, specification-based conformance, and prototype or set-cover selection are likewise established [14, 15, 16, 17, 18]. We do not propose a new oracle theory or a new set-cover algorithm. The question is whether coefficient or magnitude-response references that exactly fit one specification-certified FIR/IIR universe transfer to later catalog-blind implementations of the same masks.

This paper makes three contributions. (i) We formulate correctness as membership in a specification-defined valid set $\mathcal{V}_t$ and assemble a continuously certified FIR/IIR evaluation universe. (ii) We quantify how finite catalogs of *actual* valid realizations recover that membership on the frozen universe, including an exact observed-valid catalog-complexity diagnostic. (iii) On a catalog-blind prospective challenge of 614 independently certified realizations, we show a strong representation-dependent contrast: frozen coefficient catalogs transfer poorly, frozen magnitude-response catalogs usually transfer well, and admitting the new valids substantially expands every exact coefficient catalog.

2. SPECIFICATION SETS AND REFERENCE CATALOGS

A task t asks for an FIR impulse response or an IIR pair (b, a) . A predicate $S_t(h) \in \{0, 1\}$ encodes the registered magnitude mask and, for IIR designs, a strict pole-radius bound. Correctness is membership

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \quad (1)$$

A reference catalog $\mathcal{R}$ is a finite set of *observed* specification-valid realizations. With a distance d and a common scalar threshold τ , the acceptance set is

$$\mathcal{A}_{\tau, \mathcal{R}} = \{h : d_{\mathcal{R}}(h) \leq \tau\}, \quad d_{\mathcal{R}}(h) = \min_{r \in \mathcal{R}} d(h, r). \quad (2)$$

The reference oracle declares “correct” iff $h \in \mathcal{A}_{\tau, \mathcal{R}}$; the specification oracle declares “correct” iff $h \in \mathcal{V}_t$.

All exact-recovery statements below are restricted to an explicit finite universe $\mathcal{U}_t$. Write $V = \mathcal{V}_t \cap \mathcal{U}_t$ and $I = \mathcal{U}_t \setminus \mathcal{V}_t$, and define

$$D_V(\mathcal{R}) = \max_{v \in V} \min_{r \in \mathcal{R}} d(v, r), \quad D_I(\mathcal{R}) = \min_{i \in I} \min_{r \in \mathcal{R}} d(i, r). \quad (3)$$

Observation 1. On a frozen finite universe, a common scalar threshold recovers membership exactly if and only if $D_V(\mathcal{R}) < D_I(\mathcal{R})$. The statement is evaluation geometry, not a theorem over all filters.

The observed-valid *reference catalog complexity* K_{obs}^* is the smallest $|\mathcal{R}|$ of specification-valid occupants of V for which Observation 1 holds. Computing that finite-universe diagnostic reduces to a standard set-cover optimization [18, 17]; the optimization itself is not our contribution. K_{obs}^* measures realizable reference burden, not algorithmic novelty.

Two distances are used. Coefficient distance is a relative ℓ_2 after canonicalization (FIR: trailing-zero and sign rules; IIR: $a_0 = 1$, concatenated (b, a)). Magnitude-response distance is the RMSE of $|H|$ on the constrained pass and stop bands at $N_f = 131072$ frequency samples. Centers must be actual observed valid realizations, not arbitrary Euclidean points.

3. CERTIFIED UNIVERSE AND PROSPECTIVE PROTOCOL

The base suite comprises 20 non-unique magnitude-mask tasks: FIR low-pass, high-pass, band-pass, and band-stop at $f_s = 8$ and 16 kHz, each loose and tight (16 FIR); and IIR low-pass and high-pass at 8 kHz (4 IIR). The frozen constructed valid corpus contains 412 implementations (336 FIR, 76 IIR).

Final labels are not self-certified by the construction checker. For FIR, $P(x) = |H|^2$ with $x = \cos \omega$ is certified continuously over each constrained band by exact-rational Bernstein or Sturm sign reasoning. For IIR, we certify strict Schur stability (pole radius < 0.999) and the sign of $P_B(x) - CP_A(x)$ on every constrained band. All 412/412 constructed valids received continuous specification certification; there were no valid $\rightarrow$ invalid contradictions, and all frozen mechanism and boundary invalids certified invalid. Cosine endpoint handling is a numerical/rational certificate, not a machine-checked continuous proof. Implementation details live in the repository.

Three questions organize the audit. RQ1: when does realization-reference matching recover specification membership on this frozen universe? RQ2: how large a realizable catalog is required for exact recovery? RQ3: do exact base-fitted catalogs transfer to later implementations that satisfy the same masks?

We first froze the base catalogs and their max-safe thresholds (the largest τ that accepts no frozen base invalid). We

then protocol-locked generation across all 20 tasks and eight standard families—Remez, FIRLS, frequency sampling, and windowed FIR; Butterworth, Chebyshev I, II, and elliptic IIR—scheduled 960 attempts, continuously certified occupants, removed exact coefficient duplicates, and froze 614 new nonduplicate valid realizations (500 FIR, 114 IIR; every task; at least 13 per task) without accessing catalog identities, reference distances, transfer scores, or ambient centers. Only then did we evaluate transfer. An earlier same-order probe set participated in catalog fitting and is not this holdout. The 614 occupants are prospectively generated, catalog-blind, specification-certified realizations, not all possible filters.

4. RESULTS

4.1. Base-universe adequacy

On the frozen 412-valid universe, no single observed valid coefficient reference recovers exact membership (20/20 canonical and 20/20 best-observed non-separable). A single magnitude-response reference fails on 19/20 canonical and 18/20 best-observed tasks (Table 1). Exact observed-valid coefficient catalogs exist on 20/20 tasks, but $K^* > 10$ on every task (median 23). Library-only catalogs recover membership on 0/20 tasks in both representations. Response K^* has median 17.5 and equals 1 on two tasks.

An arbitrary Euclidean center separates 19/20 coefficient and 20/20 response base universes (one IIR low-pass task has no coefficient center), so the observed failures should not be read as intrinsic one-center inseparability; the restriction to actual realizations is consequential.

4.2. Prospective transfer

This is the primary empirical result (Fig. 1, Table 2). Frozen coefficient K_{obs}^* catalogs accepted 66/614 prospective valids (pooled 10.7%; task-macro mean 10.6%, median 4.8%; range 0–50%). Every task transferred below 75% (20/20); none reached 95%. FIR and IIR task-macro means were 11.8% and 5.5%. The same verdict is obtained under the predeclared midpoint threshold, so the coefficient failure is not an artifact of max-safe τ .

Frozen magnitude-response catalogs accepted 585/614 (pooled 95.3%; task-macro mean 94.6%, median 100%). Fifteen tasks reached at least 95%; one tight FIR band-pass task transferred at 70%. FIR and IIR macros were both about 94.6%. Response transfer is generally strong and still task-specific.

Coefficient transfer failed across all eight generator families (family rates 3.6%–18.2%); response transfer for those families ranged from 85.7% to 100%.

Enlarging the coefficient catalog on the *base* universe does not rescue prospective transfer. Canonical $K=1$, published $K \in \{3, 5\}$, and all-library coefficient oracles accepted

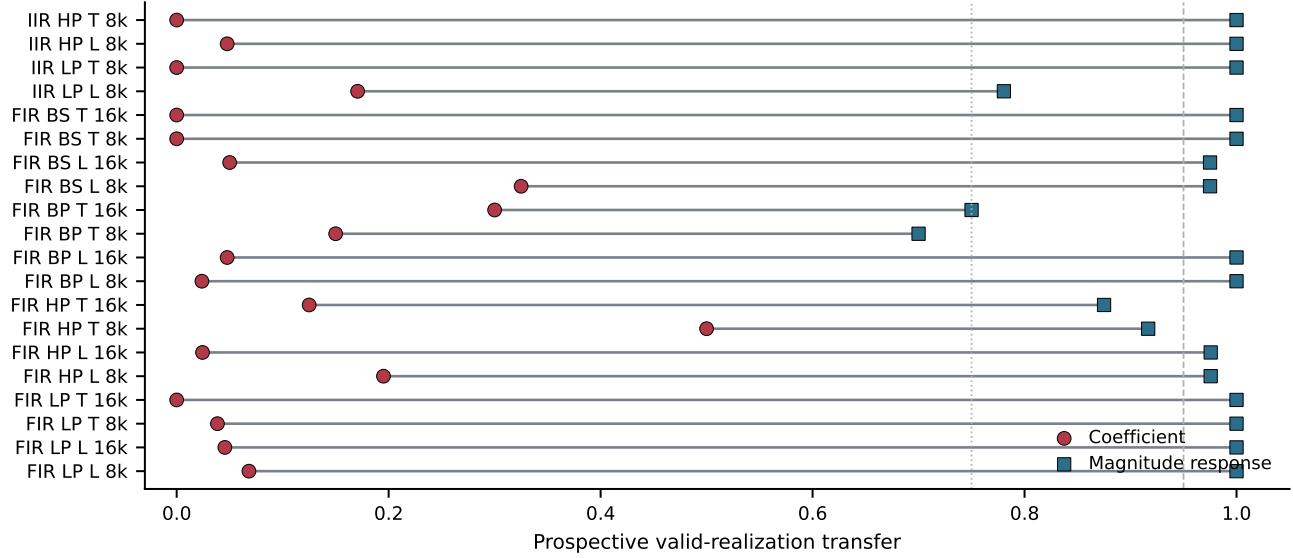


Fig. 1. Catalog-blind prospective transfer on 614 specification-certified realizations. Each row is one magnitude-mask task (L/T: loose/tight). Circles: frozen coefficient catalogs. Squares: frozen magnitude-response catalogs. Guides mark 75% and 95%.

Table 1. Exact recovery on the frozen base universe. “Non-sep.” is a finite-universe single-center failure. K^* is the minimum observed-valid catalog size.

Repr.	Can. non-sep.	Best obs.	med. K^*	Lib. exact
Coeff.	20/20	20/20	23	0/20
Resp.	19/20	18/20	17.5	0/20

Table 2. Prospective transfer of frozen base catalogs to 614 catalog-blind certified-valid realizations.

Repr.	Accepted	Pooled	Macro med.	$\geq 95\%$	$< 75\%$
Coeff.	66/614	10.7%	4.8%	0/20	20/20
Resp.	585/614	95.3%	100%	15/20	1/20

0/614; the best observed single coefficient reference accepted 7.5%; the exact K_{obs}^* catalog accepted 10.7%. The corresponding response figures are 38.9% (canonical) and 95.3% (K_{obs}^*). Increasing coefficient-catalog size enough to fit the original finite universe therefore does not produce strong prospective coefficient transfer.

4.3. Catalog maintenance

Admitting the 614 prospective valids and recomputing exact coefficient catalogs raises the median K^* from 23 to 55 (median relative growth 52.5%). Every task has $M^* > 0$: every minimum-size expanded coefficient catalog requires at least one newly admitted realization (median $M^* = 15.5$, max-

imum 37). In-sample catalog exactness therefore does not eliminate later support-expansion burden. Response catalogs grow much less (median $\Delta K = 0$; $M^* > 0$ on 9/20 tasks).

5. DISCUSSION

Coefficient matching retains realization-specific parameterization: distinct Remez, window, or elliptic occupants of the same mask can sit far apart in tap or (b, a) space, so a catalog exact on one sample can miss later valid designs (66/614). Magnitude-response matching quotients out much of that coefficient-level variability and aligns more directly with the pass/stop mask S_t , which is why the same catalogs accept 585/614 occupants. The contrast is a signal-processing representation fact, not merely that one rate is larger. Response matching remains a surrogate: it is not 100%, one task transferred below 75%, and it does not prove mask satisfaction.

A golden coefficient vector, or a finite coefficient catalog fitted to one implementation corpus, should not be read as a correctness oracle for a non-unique filter-design task without an adequacy audit. Where magnitude-response matching transfers, it can be a useful surrogate. The criterion used to define correctness in this study is still S_t .

The study is finite: 20 magnitude-mask tasks, not general DSP software, and eight standard designer families, not all implementations. A secondary invalid challenge was not balanced across all tasks; full results are in the repository. No infinite-universe claim is made, and magnitude-response matching is not asserted to be sound or complete in general.

6. CONCLUSION

Reference matching should be audited as a surrogate for a digital-filter specification, not assumed equivalent to it. Across 614 prospectively generated valid FIR/IIR realizations, coefficient catalogs that were exact on one implementation universe transferred poorly, while magnitude-response catalogs were substantially more stable. Correctness evaluation should distinguish specification compliance from realization identity and choose the reference representation accordingly.

Code and frozen artifacts are in the accompanying public repository.

Compliance with ethical standards. No funding was received for conducting this study. The authors have no relevant financial or nonfinancial interests to disclose. The study did not involve human participants, animal subjects, or personally identifiable data.

7. REFERENCES

- [1] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, Prentice Hall, Upper Saddle River, NJ, 3 edition, 2010.
- [2] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, Pearson Prentice Hall, Upper Saddle River, NJ, 4 edition, 2007.
- [3] L. R. Rabiner and B. Gold, *Theory and Application of Digital Signal Processing*, Prentice Hall, Englewood Cliffs, NJ, 1975.
- [4] J. F. Kaiser, "Nonrecursive digital filter design using the I_0 -sinh window function," in *Proc. IEEE Int. Symp. Circuits and Systems*, 1974, pp. 20–23.
- [5] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, Jan. 1978.
- [6] L. R. Rabiner, B. Gold, and C. A. McGonegal, "An approach to the approximation problem for nonrecursive digital filters," *IEEE Trans. Audio and Electroacoustics*, vol. AU-18, no. 2, pp. 83–106, June 1970.
- [7] T. W. Parks and J. H. McClellan, "Chebyshev approximation for nonrecursive digital filters with linear phase," *IEEE Trans. Circuit Theory*, vol. CT-19, no. 2, pp. 189–194, Mar. 1972.
- [8] J. H. McClellan, T. W. Parks, and L. R. Rabiner, "A computer program for designing optimum FIR linear phase digital filters," *IEEE Trans. Audio and Electroacoustics*, vol. AU-21, no. 6, pp. 506–526, Dec. 1973.
- [9] L. R. Rabiner, J. H. McClellan, and T. W. Parks, "FIR digital filter design techniques using weighted Chebyshev approximation," *Proc. IEEE*, vol. 63, no. 4, pp. 595–610, Apr. 1975.
- [10] S. Butterworth, "On the theory of filter amplifiers," *Experimental Wireless and the Wireless Engineer*, vol. 7, pp. 536–541, 1930.
- [11] A. G. Constantinides, "Spectral transformations for digital filters," *Proc. IEE*, vol. 117, no. 8, pp. 1585–1590, Aug. 1970.
- [12] L. B. Jackson, *Digital Filters and Signal Processing*, Springer, Boston, MA, 3 edition, 1996.
- [13] A. Antoniou, *Digital Signal Processing: Signals, Systems, and Filters*, McGraw-Hill, New York, NY, 2006.
- [14] E. J. Weyuker, "On testing non-testable programs," *The Computer Journal*, vol. 25, no. 4, pp. 465–470, 1982.
- [15] E. T. Barr, M. Harman, P. McMinn, M. Shahbaz, and S. Yoo, "The oracle problem in software testing: A survey," *IEEE Trans. Software Eng.*, vol. 41, no. 5, pp. 507–525, May 2015.
- [16] T. Huuhtanen, J. Itkonen, and C. Lassenius, "Combining algebraic and domain testing to design adequate test cases for signal processing algorithms," in *Proc. IEEE 8th Int. Conf. Software Testing, Verification and Validation (ICST)*, 2015, pp. 1–10.
- [17] P. E. Hart, "The condensed nearest neighbor rule," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 3, pp. 515–516, May 1968.
- [18] V. Chvátal, "A greedy heuristic for the set-covering problem," *Math. Oper. Res.*, vol. 4, no. 3, pp. 233–235, Aug. 1979.